

CAIKA-RAG-Typhoon Datasheet

Overview

CAIKA-RAG-Typhoon (Corporate AI Knowledge Assistant) is a retrieval-augmented generation (RAG) system designed to provide accurate, contextual responses to queries about corporate knowledge using local document repositories.

Developer Information

- **Developer:** Thapagorn Hanrak
- **Model:** scb10x_llama3.2-typhoon2-t1-3b-research-preview-gguf

System Architecture

Core Components

1. Document Processing Pipeline

- Loader: Uses PyPDFLoader to process PDF documents
- Text Splitting: RecursiveCharacterTextSplitter with 1000-character chunks and 200-character overlap
- Storage: Documents stored in `data/documents` directory

2. Embedding System

- Model: HuggingFaceEmbeddings (default: "scb10x/llama3.2-typhoon2-t1-3b-research-preview")
- Vector Store: FAISS for efficient similarity search
- Storage: Vector embeddings stored in `data/vector_store`

3. Language Model

- Backend: LlamaCpp for optimized local inference
- Model: "scb10x_llama3.2-typhoon2-t1-3b-research-preview-gguf"
- Configuration: GPU acceleration enabled with half-precision (f16_kv) for efficient memory usage

4. RAG Pipeline

- Retriever: FAISS-based similarity search
- Default Settings: k=4 documents retrieved per query

- Chain Type: "stuff" (combines all relevant documents into a single context)

Technical Specifications

Performance Optimization

- Optimized for Apple Silicon (M1) with Metal Performance Shaders support
- Reduced context window (1024 tokens) to conserve memory
- Half-precision key/value cache for optimized performance
- Multi-threaded execution (4 threads default)

User Interface

- Built with Streamlit for interactive web-based access
- Features:
 - Document upload and processing
 - Model selection and parameter adjustment
 - Conversation history tracking
 - Source document visibility for transparency

Dependencies

- langchain & langchain_community
- sentence-transformers
- faiss-cpu
- pypdf
- tiktoken
- llama-cpp-python
- streamlit

Usage Parameters

LLM Generation Parameters

- Temperature: 0.7 (default, adjustable via UI)
- Top-p: 0.95 (default, adjustable via UI)
- Max Tokens: 1024 (default, adjustable via UI)

Document Retrieval Parameters

- Retrieved Documents: 4 (default)
- Adjustable retrieval parameters:
 - k: Number of documents to retrieve

- `fetch_k`: Number to initially fetch before filtering
- `score_threshold`: Minimum similarity score

Implementation Requirements

System Requirements

- MacOS with Apple Silicon (M1/M2) preferred
- Minimum 8GB RAM recommended
- 1GB+ disk space for model and vector store

Directory Structure

- `models/`: Storage for LLM model files
- `data/documents/`: PDF document storage
- `data/vector_store/`: Generated vector embeddings
- `logs/`: Application logs

Operational Notes

- The system must be initialized before use via the "Initialize CAIKA" button
- Documents must be processed before queries can be answered
- Response quality depends on the quality and coverage of the supplied documents
- All processing happens locally, ensuring data privacy